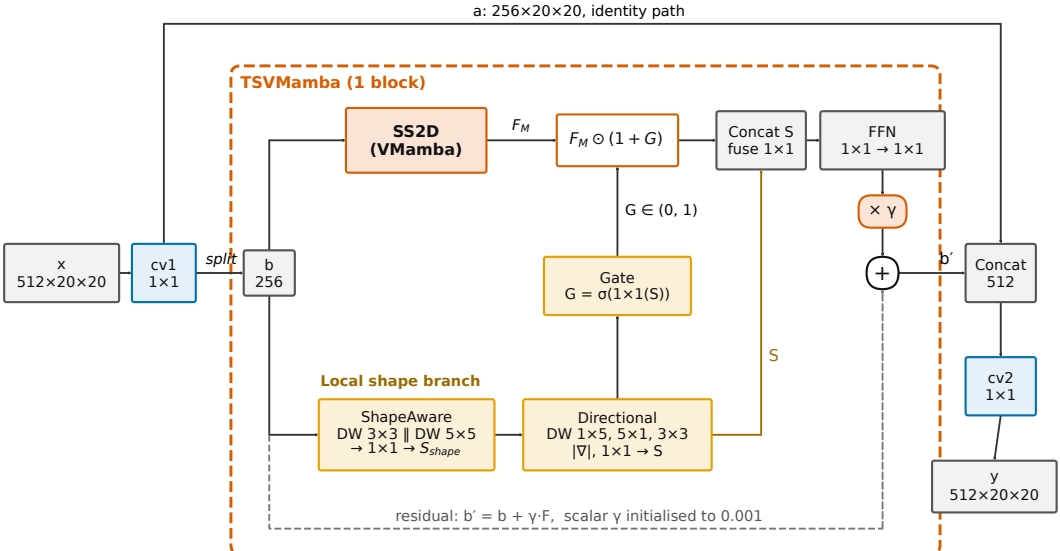
a: 256×20×20, identity path
TSVMamba (1 block)
x
512×20×20
cv1
1×1
split
b
256
SS2D
(VMamba)
$F_M$
$F_M \odot (1+G)$
Concat S
fuse 1×1
FFN
1×1 → 1×1
× γ
G ∈ (0, 1)
Gate
G = σ(1×1(S))
S
b′
Concat
512
cv2
1×1
y
512×20×20
Local shape branch
ShapeAware
DW 3×3 ‖ DW 5×5
→ 1×1 → $S_{shape}$
Directional
DW 1×5, 5×1, 3×3
|∇|, 1×1 → S
residual: b′ = b + γ·F, scalar γ initialised to 0.001